Hotel Booking Cancellation Analysis

Business Problem

In recent years, City Hotel and Resort Hotel have experienced high cancellation rates. As a result, both hotels face several issues, including reduced revenue and inefficient use of hotel rooms. Therefore, reducing cancellation rates has become the primary goal for both hotels in order to improve revenue generation and operational efficiency.

This report focuses on analyzing hotel booking cancellations and identifying the factors that influence them. The analysis will also help provide insights that can assist hotels in improving their strategies and decision-making.

Assumptions

1. No unusual occurrences between 2015 and 2017 have a significant impact on the dataset used for the analysis.
2. The information available in the dataset is still relevant and can be used to analyze hotel business strategies effectively.
3. There are no unexpected negative effects on hotels when implementing the recommended techniques.
4. The hotels are not currently using any of the suggested solutions proposed in this analysis.
5. Booking cancellations are the biggest factor affecting the hotel's revenue generation.
6. Cancellations lead to vacant rooms during the reserved booking period.
7. Clients usually make reservations and cancellations within the same year.

Research Questions

1. What factors influence hotel reservation cancellations?
2. How can hotel reservation cancellations be reduced?
3. How can the analysis help hotels make better pricing and promotional decisions?

Hypothesis

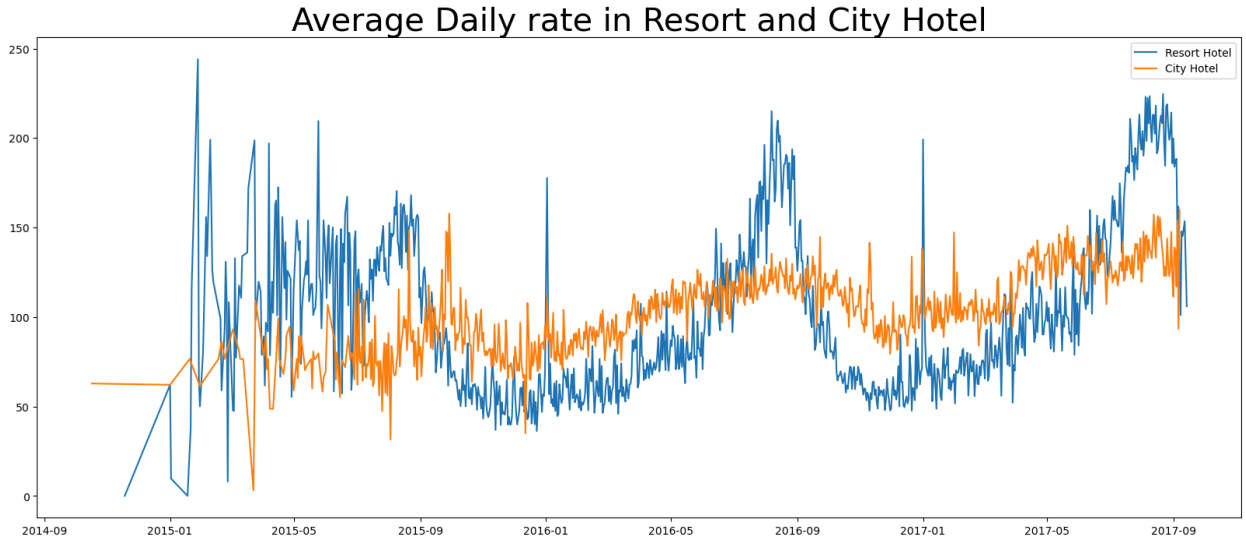
1. Higher room prices lead to more booking cancellations.

2. When the waiting list for reservations is longer, customers are more likely to cancel their bookings.
3. A large number of customers make reservations through offline travel agents.

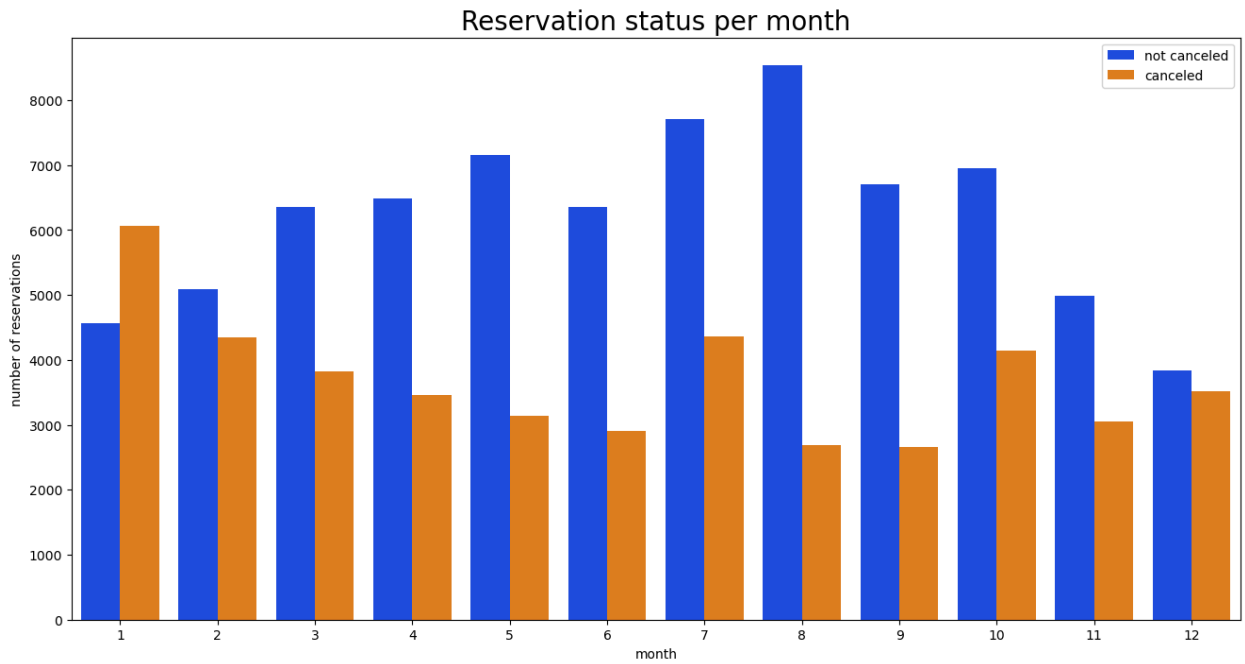
Analysis and Findings

The accompanying bar graph shows the percentage of reservations that are canceled and those that are not. It is obvious that there are still a significant number of reservations that have not been canceled. There are still 37% of clients who canceled their reservation, which has a significant impact on the hotels' earnings.

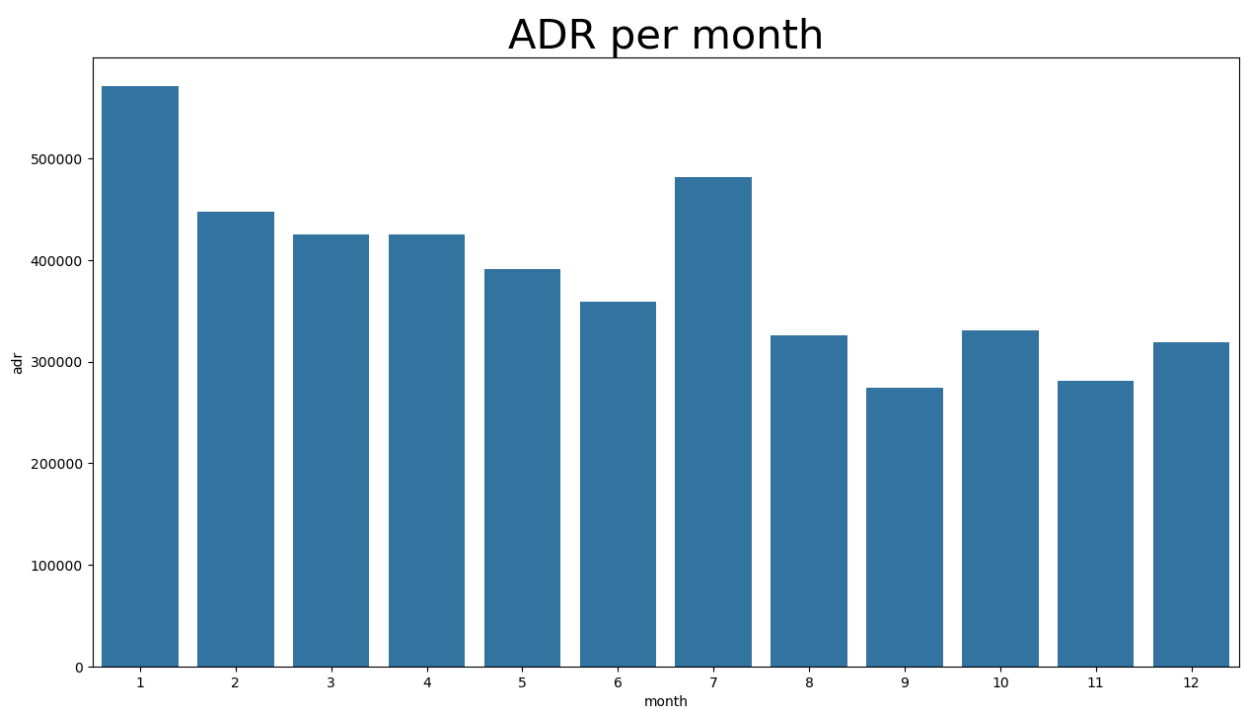
In comparison to resort hotels, city hotels have more bookings. It’s possible that city hotels are more expensive than those in cities.



The line graph above shows that, on certain days, the average daily rate for a city hotel is less than that of a resort hotel, and on other days, it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.

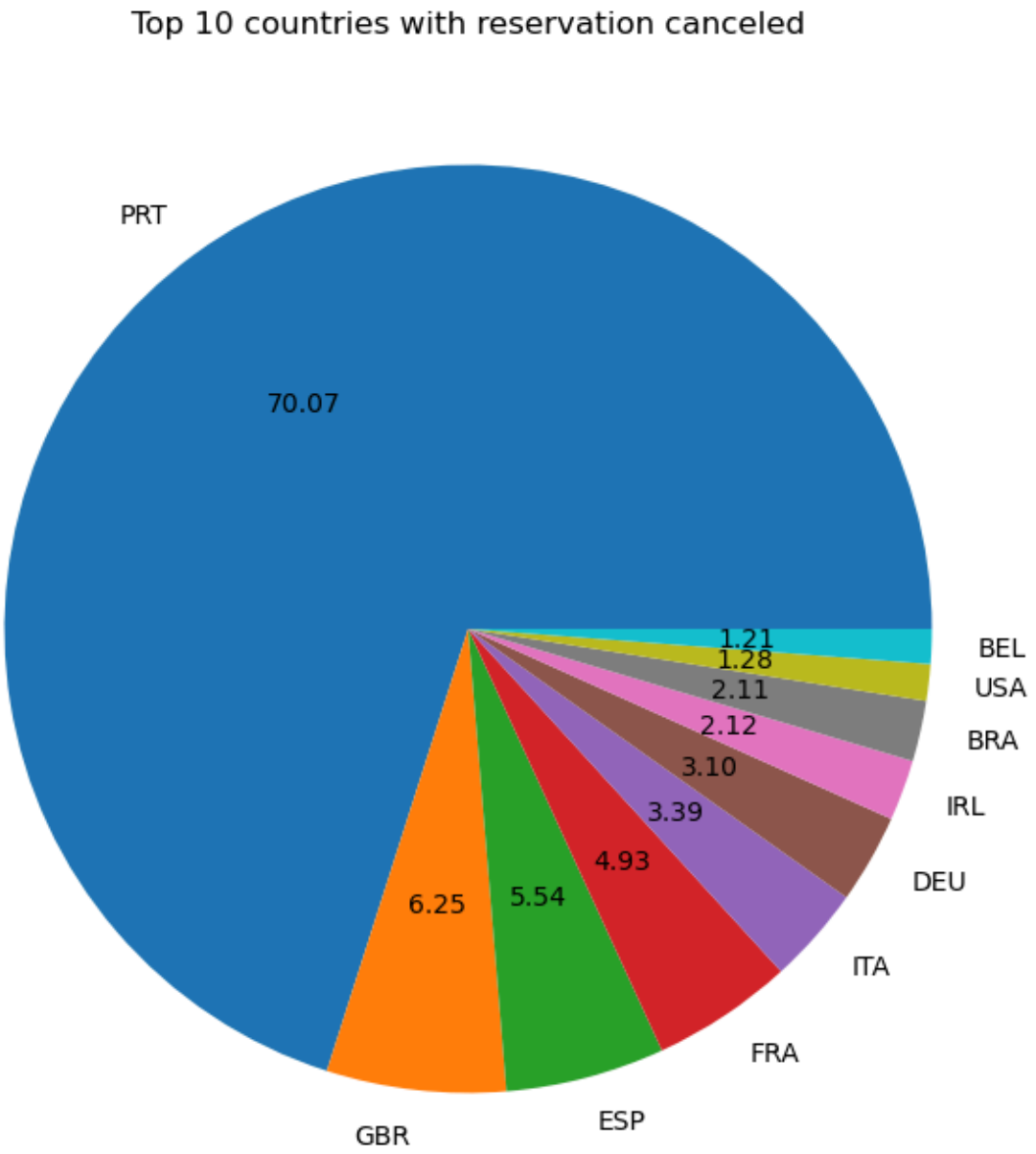


We have developed the grouped bar to analyze the months with the highest and lowest reservation levels according to reservation status. As can be seen, both the number of confirmed reservations and the number of canceled reservations are largest in the month of August, whereas January is the month with the most canceled reservations.

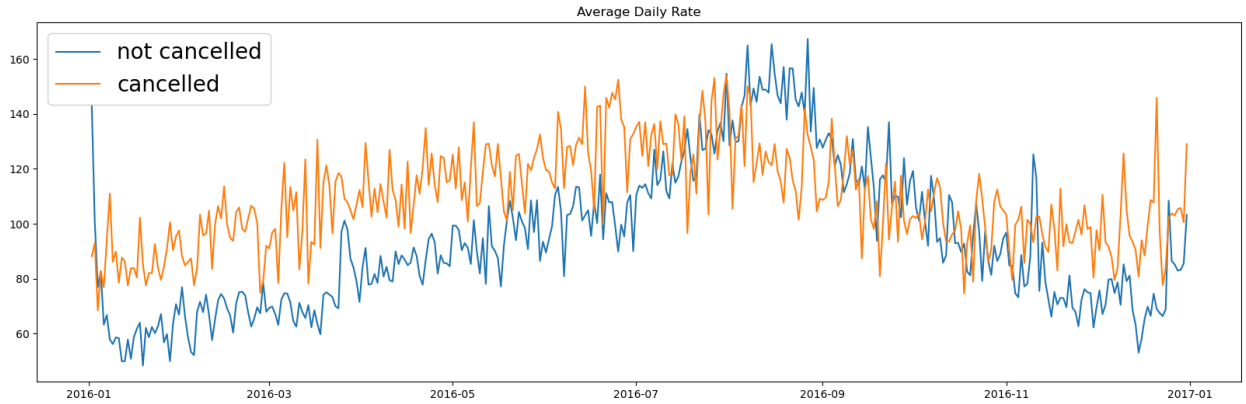


The bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is solely responsible for the cancellation.

Now, let’s see which country has the highest reservation canceled. The top country is Portugal with the highest number of cancellations.



Let's check the area from where guests are visiting the hotels and making reservations. Is it coming from Direct or Groups, Online or Offline Travel Agents? Around 46% of the clients come from online travel agencies, whereas 27% come from groups. Only 4% of clients book hotels directly by visiting them and making reservations.



As seen in the graph, reservations are canceled when the average daily rate is higher than when it is not canceled. It clearly proves all the above analysis, that the higher price leads to higher cancellation.

Suggestions

1. Cancellation rates rise as the price does. In order to prevent cancellations of reservations, hotels could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to the consumers.

2. As the ratio of the cancellation and not cancellation of the resort hotel is higher in the resort hotel than the city hotels. So the hotels should provide a reasonable discount on the room prices on weekends or on holidays.

3. In the month of January, hotels can start campaigns or marketing with a reasonable amount to increase their revenue as the cancellation is the highest in this month.

4. They can also increase the quality of their hotels and their services mainly in Portugal to reduce the cancellation rate.